Tahlia Wu

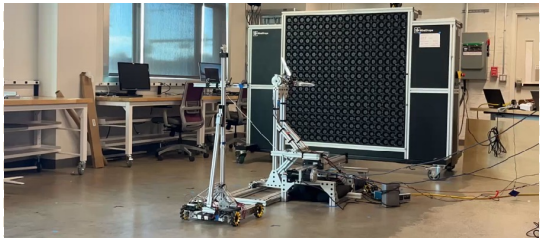
Mechanical Engineering & Robotics @ UPenn 2028

Skilled in both software and hardware, I specialize in electromechanical system integration and automating data analysis.



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Engineering Projects



Robotics

FlowBot



Precision Machining

Kirby Stirling Engine



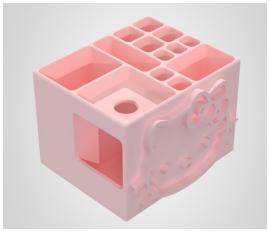
Product Design

Floating Shelf



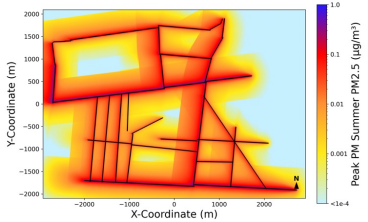
Product Design

Light Box Shelf



Product Design

Makeup Organizer



Data Analysis

Air Pollution Model

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Mechatronics

FlowBot

Embedded Systems

PID Control

FEA Analysis

DAQ

Python

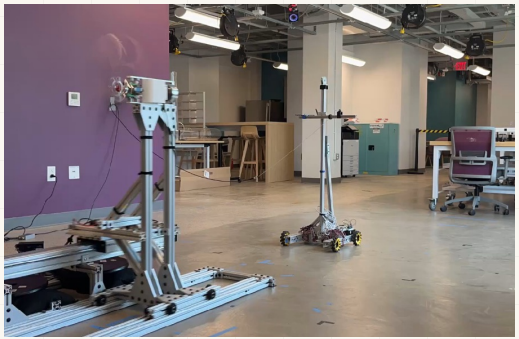
Electromechanical Integration

End-to-end engineered a mechatronic system with minimal oversight

- Designed a **vertical linear actuator with 80/20 from scratch** for <$200
- Designed in CAD and **3D-printed** hardware integration components
- **PID control loop** with OptiTrack motion capture achieveing 6mm accuracy
- Diagnosed and debugged a **data buffer overflow error** in software







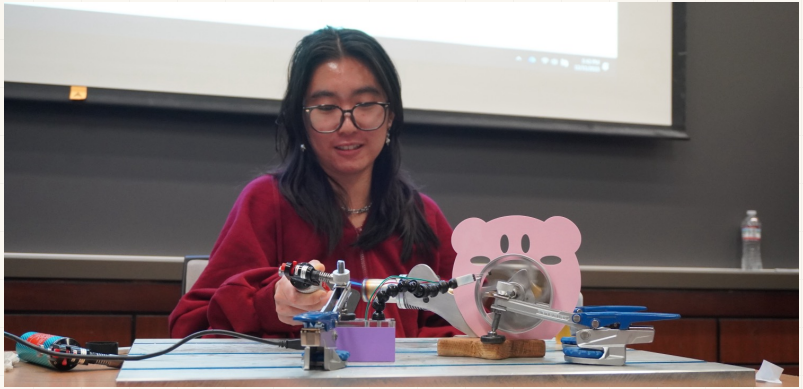
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Machining

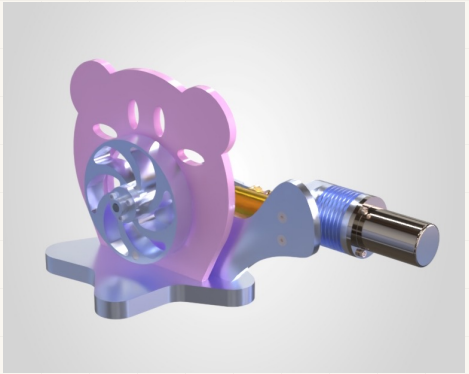
Kirby Stirling Engine

CNC / Mastercam | Lathe | 2.5-Axis Mill

SolidWorks / CAD | GD&T | Tolerance Stack Up



Precision-machined 17 parts from stock to build a fully functional Stirling engine, showcasing **end-to-end product development** from CAD/CAM integration (SolidWorks, Mastercam).





Materials machined:

- C360 Brass
- 1020 Carbon Steel
- 6061 Aluminum
- 303 Stainless Steel
- Wood
- Acrylic

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Product Design

Floating Shelf

Laser Cutting | SolidWorks / CAD | DFM/DFA | Rapid Prototyping

Driven by the need to optimize spatial efficiency, I engineered a minimalist display solution featuring a modular architecture for seamless **Design for Disassembly (DfD).**

The system was modeled in SolidWorks and **rapidly prototyped** with **laser-cut** 0.25" clear acrylic to maintain an unobtrusive visual footprint.





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Product Design

Light Box Shelf

Laser Cutting | SolidWorks / CAD | Joint Design | DFM/DFA | Rapid Prototyping

Merging form and function, I engineered a figurine shelf that seamlessly **integrates an aesthetic display with practical storage**.

The concept was parametrically modeled in **SolidWorks** and rapidly prototyped using laser-cut ¼” MDF.





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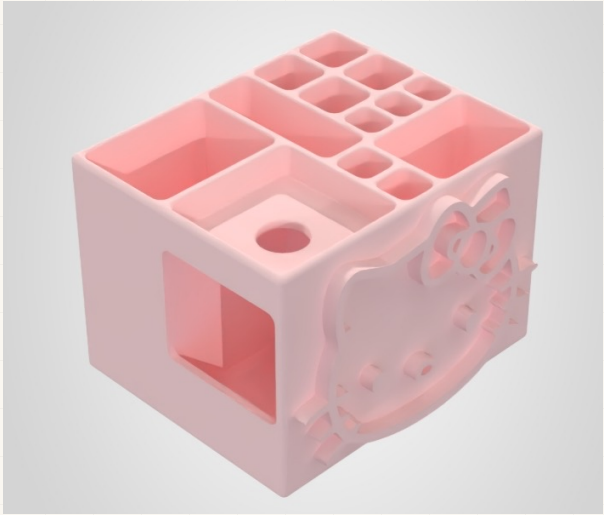
Product Design

Hello Kitty Makeup Organizer

3D Printing | SolidWorks / CAD | DFM/DFA | Rapid Prototyping

After noticing my makeup frequently falling off due to my limited counter space, I measured my actual makeup collection and used **SolidWorks to design** a vertical organizer perfectly sized for my needs.

I 3D printed this using ABS with dissolvable supports and finished with a personalized spray-painted color.



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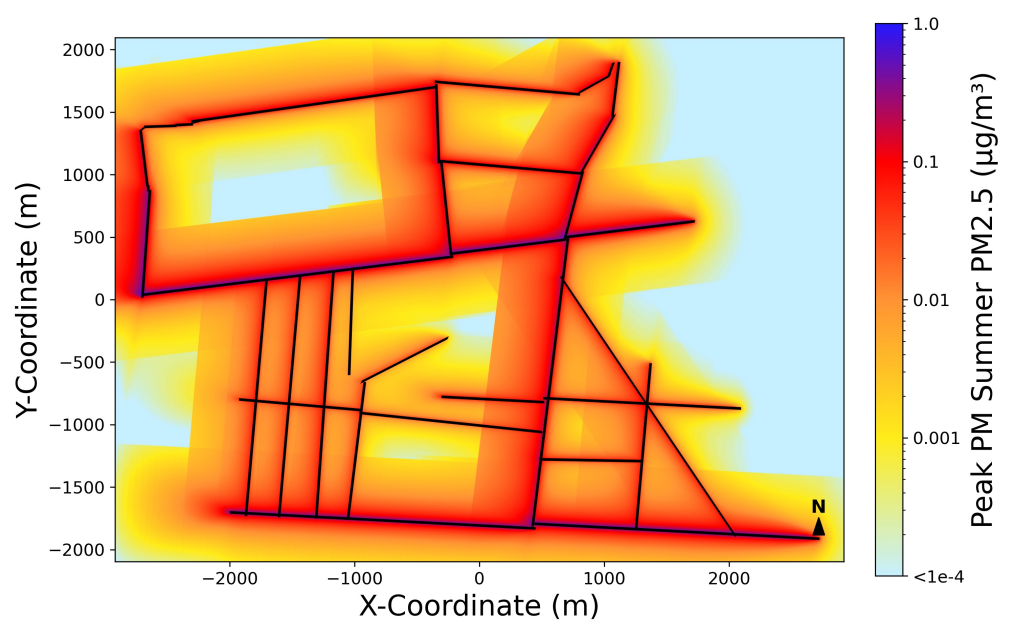
Data Analysis

Air Pollution Model

R | EPA MOVES/R-LINE

Data Analysis | Virtual Simulation

To overcome **computational bottlenecks** in high-resolution vehicular emissions modeling, I engineered a scalable **superposition algorithm and unit-road dictionary,** successfully validating this accelerated pipeline on a 36-road network in Ohio.



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Get in Touch

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